



Effects of microplastic exposure on the blood biochemical parameters in the pond turtle (*Emys orbicularis*)

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Abstract

The accumulation of microplastics (MPs) is a growing problem in aquatic ecosystems. Despite increased research on MPs in the last decade, their potential threat to freshwater ecosystems remains an open question. In the present study, the negative impacts of MPs were investigated on blood biochemical parameters in the European pond turtle (*Emys orbicularis*). Pond turtles were distributed into three experimental groups ($n = 9$ for each group) and were fed diets containing 250, 500, and 1000 mg MPs (PE100 polyethylene) per kg of food for 30 days, and a control group fed with a standard uncontaminated diet. The results indicated that exposure to 500 and 1000 mg kg⁻¹ MPs caused a significant increase in the activities of alanine and aspartate aminotransferases, and in the levels of cholesterol, glucose, creatinine, urea, and calcium (Ca⁺²) compared with the control group. On the contrary, the activity of gamma-glutamyl transferase and the levels of total protein, albumin, total immunoglobulins, and phosphorus were significantly reduced in *E. orbicularis* exposed to 500 and 1000 mg kg⁻¹ MPs when compared with the controls. In all the MP-exposed groups, the activity of lactate dehydrogenase and globulin and magnesium (Mg⁺²) levels were significantly reduced; while creatine phosphokinase and alkaline phosphatase activities were increased with respect to the control turtles. A significant decrease in triglyceride levels was reported in *E. orbicularis* exposed to 1000 mg kg⁻¹ MPs. MPs intake induced notable alterations in blood biochemical parameters of *E. orbicularis*. These results suggest that changes in the blood biochemical parameters could be an appropriate bio-indicator to evidence the existence of tissue damage in *E. orbicularis*.

Keywords Polyethylene · *Emys orbicularis* · Biomarkers · Freshwater

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Introduction

In recent years, plastic debris has turned one of the major environmental challenges worldwide (Avio et al. 2017; Alimba and Faggio 2019; Ajith et al. 2020; Freeman et al. 2020). Microplastics (MPs) are small fragments of plastics less than 5 mm in size that can be released directly into the environment or derived from the progressive fragmentation of larger elements (Zhang et al. 2020). Clothing and textile industries, cosmetics, raw materials used in the plastics industry, as well as fishing activities are the main point and non-point sources of contaminating aquatic ecosystems with MPs and nanoplastics (NPs) (Clukey et al. 2017; De Frond et al. 2019). Eventually, these materials enter surface water through urban wastewater and surface runoff (Alimba and Faggio 2019). A large volume of scrubbers drains surface water through domestic wastewater every year (Nematdoost Haghi and Banaee 2017; Auta et al. 2017). Larger plastic can be broken down into smaller particles by physical erosion and ultraviolet (UV)

rays and end up in water ecosystems through surface runoff (Wright et al. 2013; Selvam et al. 2021; Wahl et al. 2021).

In Iran, 350,000–400,000 t of plastic materials are produced annually (in addition to the imported plastic products), most of which are discarded after use. However, there is no official report on the level of MPs in the freshwater ecosystems in Iran. Dehghani et al. (2017) reported that more than 605 MPs (250–500 μm) were found in 30 g of dust on the streets of Tehran. These MPs eventually enter surface and groundwater through surface runoff. Furthermore, in Tehran, around 1 t of plastic bags and disposable containers that are non-recyclable are discarded daily, a significant part of which reaches aquatic ecosystems (Dehghani et al. 2017). Wang et al. (2020) revealed that the volume of MPs in the effluent from treatment plants could be more than 9×10^4 particles per m^{-3} . A study was done by Sun et al. (2019) showed that more than 2×10^6 MPs particles could discharge daily into freshwater ecosystems from wastewater treatment plants. Scholars reported concentrations of MPs in both the sediment and water of Brisbane River in Australia (0.18 to 129.2 mg kg^{-1}) (He et al. 2020), the Nakdong River in South Korea (1970 ± 62 particles kg^{-1}) (Eo et al. 2019), urban recipient in Norway (12,000–200,000 items kg^{-1}) (Haave et al. 2019), and in an urban lake of London (539 items kg^{-1}) (Turner et al. 2019). Therefore, these substances quickly become threatening agents for aquatic organisms (Browne et al. 2011; Browne 2015; Chen et al. 2020; Murano et al. 2020). Plastic materials directly or indirectly cause damage and jeopardize the survival of a wide variety of aquatic organisms (Banaee et al. 2019; Barboza et al. 2018a; Batel et al. 2018; Guzzetti et al. 2018). In fact, plastic litter can cause death by blocking of the respiratory or gastrointestinal tract (Caron et al. 2018; Ašmonaitė et al. 2018; Wen et al. 2018b; Prokić et al. 2019; Savoca et al. 2019; Roch et al. 2019). The bioaccumulation of MPs and NPs in the edible aquatic organism can increase the risk of their transmission to humans through the food chain (Tosetto et al. 2017). Furthermore, plastic structures can provide a good basis for the growth and development of pathogenic or invasive organisms, favoring their dispersal (Wu et al. 2019). Plastic debris can adsorb a large variety of compounds on their surface, which can cause a synergistic or additive effect on the toxicity of these chemical pollutants (Banaee et al. 2019; Barboza et al. 2018b; Batel et al. 2018; Wen et al. 2018a; Zhang et al. 2019). Thereby, plastic materials may act as vectors highly capable of transmitting different types of biological and chemical contaminants (Rainieri et al. 2018; Zhang et al. 2019). Consequently, assessing the impact of MPs on water ecosystems and different aquatic species is very important.

Although freshwater ecosystems are one of the primary sources of contamination with MPs (Strungaru et al. 2018; Adeogun et al. 2020; Fu and Wang 2019; Meng et al. 2020), there is not much data on the effects of MPs on freshwater organisms compared with marine organisms (Dawson et al.

2018; Foley et al. 2018; Lusher et al. 2018; Adeogun et al. 2020). MPs travel through the food chain and may increase the bioaccumulation of these contaminants in predators (Nelms et al. 2018). Duncan et al. (2019) investigated the presence of MPs in the digestive system of seven species of marine turtles from the Atlantic Ocean, Pacific Ocean, and the Mediterranean Sea, evidencing considerable amounts of MPs in all species, mainly blue or black plastic fibers. Caron et al. (2018) identified diverse concentrations of polyethylene, high-density polyethylene, polyvinyl chloride, polypropylene, and polystyrene (aminoethyl) less than 100 μm in the digestive system of the green sea turtle (*Chelonia mydas*). The most common plastics in the digestive system of loggerhead turtles (*Caretta caretta*) captured from the North Atlantic subtropical gyre, in the Azores region, were polyethylene and polypropylene fragments (Pham et al. 2017). The presence of MPs in the digestive system of green turtles, olive ridley, loggerhead, and leatherback sea turtles have also been recorded (Clukey et al. 2017; Jung et al. 2018; Duncan et al. 2019). Besides, many turtles living in freshwater environments are predators and omnivores that at high risk of being affected by the intake and bioaccumulation of MPs (Abreo et al. 2016). Some of the negative consequences of MP intake are oxidative stress (Alomar et al. 2017), genotoxicity (Bacolod et al. 2017; Limonta et al. 2019), disturbance in the expression of genes involved in cell signaling, and antioxidant defense system enzymes (Rainieri et al. 2018; Yu et al. 2018). Furthermore, MPs could induced changes in blood parameters (Nematdoost Haghi and Banaee 2017; Banaee et al. 2019) and weaken the immune system (Banaee et al. 2019; Limonta et al. 2019; Murano et al. 2020). A reduced appetite (Cole et al. 2015; Critchell and Hoogenboom 2018), imbalance in the energy budget (Watts et al. 2014), disturbance in energy production process in mitochondria (Cui et al. 2017; Pitt et al. 2018; Kim et al. 2019), a significant reduction in fecundity (Sussarellu et al. 2016), and a lower survival rate (Oliveira et al. 2013) were also observed in the aquatic organisms exposed to MPs. These physiological, biochemical, and genetic disorders might be due to the release of toxic and chemical components of plastic material (O'Hara et al. 2019; Yan et al. 2020). However, the potential toxic effects of MPs on turtles are still poorly understood.

Therefore, in order to fill this information gap, the European pond turtle (*Emys orbicularis*) was selected as an experimental model. *E. orbicularis* is mainly found in wetlands of southern and central Europe, north Africa (Morocco, Algeria, and Tunisia), and in the west and central Asia including Iran (Rossler 2000a; Rossler 2000b; Meeske 2000; Bringsoe 2001; César and Rivera 2001; Zuffi et al. 2006; Ayres and Cordero 2007; Heiko et al. 2014; Shayegh et al. 2016; Akman et al. 2020). The pond turtle is an omnivorous species that feeds on aquatic insect larvae, fish larvae, and aquatic plants. Unfortunately, in Iran, little attention is paid

to the conservation of this species, and its population is endangered in some areas of the country. MPs may threaten the survival of this endangered species of the turtle. This study hypothesizes that intake of a diet containing MPs for 30 days can significantly affect the blood biochemical parameters in *E. orbicularis*.

Materials and methods

Chemical material

High-density polyethylene (PE100) powder (< 0.2 mm) was purchased from Eshragh Trading Co., Iran. All diagnostic kits used to measure biochemical parameters were obtained from Pars Azmun Co., Iran.

Animals

Turtles were caught from the margins of local aquatic ecosystems. The turtles were transferred to the laboratory in a secure plastic container with a damp padded substrate (e.g., a towel). After rinsing, their bodies were disinfected with brine and short-term formalin (10%) bath (200 mg L⁻¹). Turtles were maintained in methacrylate tanks with dechlorinated tap water and were adapted to the laboratory conditions for a month (temperature, 22 ± 2 °C; photoperiod cycle, 14 light, 10 dark; dissolved oxygen, 8.1 ± 0.7 mg L⁻¹; pH, 7.3 ± 0.4; salinity, 0.3 ± 0.02 g L⁻¹; and electrical conductivity, 693.83 ± 76 µS cm⁻¹). During this period, the animals were fed a standard formulated control diet (Table 1) as well as fresh vegetables (lettuce and cucumber). The turtle maintenance conditions and the experimental procedure followed the guidelines and were approved by the Ethics Committee of the Faculty of Veterinary Medicine, Shiraz University, Iran (SURC973/46-2679-2).

Table 1 Chemical composition of experimental diets (g. 100 g⁻¹ on dry matter basis)

Diets	
Size	4.5–5 mm
Crude protein	43%
Crude lipid	15%
Crude fiber	2.5%
Moisture	10%
Digestible energy	4400 Kcal kg ⁻¹
Phosphorus available	0.7%
Total calcium	2.2%
Lysine	5.9%
Methionine	1.9%
Threonine	3.2%
Total ash	8.5%

Experimental diet preparation

MPs were incorporated in different concentrations (0, 250, 500, and 1000 mg kg⁻¹) to the diet following previously described procedures (Espinosa et al. 2017; Hantoro et al. 2019). Since there are no studies on the presence of MPs in freshwater ecosystems in Iran, values similar to those found in coastal areas of the country have been selected (Akhbarizadeh et al. 2017; Naji et al. 2017). The ingredients of the basal diet and the high-density polyethylene (PE100) powder were mixed in appropriate concentrations to reach four experimental diets with the following amounts of MPs: 0 mg (control group), 250, 500, and 1000 mg kg⁻¹ MPs. Each diet was blended with distilled water to obtain a homogenous paste. Then, this dough was mixed with the mixer and passed through a meat grinder. Subsequently, this paste was dried by warm air for 24 h and then broken into pellets of approximately 0.5 mm long. The pellets were packed and stored at 4 °C until use.

Experimental design

Thirty-six turtles with 306.6 ± 54.9 g average weight and 123.5 ± 36.2 mm average length were distributed in four tanks (0.75 × 1.5 × 1 m³) containing 50 L of water (9 turtles in each group). The number of biological samples was determined based on the resource equation method (Charan and Kantharia 2013). Group I as a control group was fed with a control and un-polluted diet. Turtles of groups II, III, and IV were fed with contaminated food with 250, 500, and 1000 mg kg⁻¹, respectively, for 30 days (Table 1). In the trial days, turtles were kept in the same laboratory conditions as the adaptation period. Every 24 h, the water was exchanged and leftover food, and the fecal matter was removed by siphoning the aquarium floor. During the experimental procedure, animals were fed with the control or experimental diets twice a day, and feeding was stopped 24 h before sampling.

At the end of the experimental procedure, the turtles were anesthetized with an isoflurane-impregnated cotton cloth in a sealed plastic box to prevent their pedal reflexes during blood sampling (Filogonio and Crossley II 2019). Then, blood was obtained from the sub-carpacal vein or sinus using a 22 ga. needle and 3-mL syringe without anticoagulant. Sampled blood was immediately transferred into a 1.5mL micro-centrifuge tubes and, after coagulation, centrifuged for 15 min at 6000 g at 4 °C to separate the blood cells from the serum. The serum was then recovered and divided into three different aliquots. All serum samples were stored at - 25 °C until biochemical analyses (< 6 days).

Biochemical analysis

The activities of all the following parameters were determined using commercial kits (Pars Azmun Co., Iran) in an UV/Vis

spectrometer (UNICO Model 2100). The activities of alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were measured at 340 nm, and pyruvate/NADH and oxaloacetate/NADH were used as substrates, respectively (Moss and Henderson 1999). Alkaline phosphatase (ALP) activity was determined using P-nitrophenyl phosphate (pNPP) and monitored at 450 nm (Moss and Henderson 1999). Lactate dehydrogenase (LDH) activity was measured at 340 nm using NADH/pyruvate as substrates (Moss and Henderson 1999). Gamma-glutamyl transferase (GGT) was evaluated following the method of Moss and Henderson (1999) at 405 nm using L- γ -glutamyl-3-carboxy-4-nitroanilide and glycylglycine. Creatine phosphokinase (CPK) activity was detected at 340 nm according to the protocol described by Moss and Henderson (1999), using creatinine phosphate as substrate.

Total protein level was estimated using Biuret reaction at 545 nm. Albumin levels were measured using Pars Azmun kits and applying bromocresol green method at 545 nm (Johnson et al. 1999). Glucose level was measured following the method developed by Sacks (1999) using glucose oxidase at 546 nm. Cholesterol and triglyceride levels were measured at 546 nm using the method of Rifai et al. (1999) using cholesterol oxidase (CHOD)/peroxidase (POD) and glycerol phosphate oxidase (GPO) after hydrolysis by lipoprotein lipase, respectively. The levels of serum creatinine were estimated at 495 nm using the Jaffe method based on the reaction of creatinine with picric acid in an alkaline environment (Foster-Swanson et al. 1994).

Urea concentration in serum was determined at 578 nm, using urea decomposition rate by urease to produce ammonia and carbon dioxide. The levels of calcium ions were measured in a neutral environment after reacting with Arsenazo III to form an intense purple-colored complex and monitored at 630 nm (Endres and Rude 1999). Magnesium ions were determined in an alkaline environment after reacting with xylidyl blue, producing a violet complex read at 546 nm (Endres and Rude 1999). Phosphorus ions were measured by reacting with ammonium molybdate in the presence of sulfuric acid and monitored at 340 nm (Endres and Rude 1999).

Statistical analysis

The Shapiro-Wilk test was used to assess data normality. One-way analysis of variance (ANOVA) was done to determine the existence of statistical differences between blood biochemical parameters. Statistical differences among groups were analyzed using the Duncan test. Differences between treatments and times were considered significant at P value < 0.05 . The results are presented as mean $\pm$ standard deviation of the mean (SD). Statistical analyses were carried out using the GraphPad Prism (8.0.2).

Results and discussion

In the present study, no significant changes were observed in the feeding status of turtles treated with different concentrations of MPs when compared with the control group. In fact, the turtles completely ingested all the food provided daily without differences between groups. The absence of differences in the intake indicates that the changes in the biochemical parameters would derive from the presence of MPs and not from differences in the food intake. In addition, throughout the experimental procedure, no unusual behaviors were observed in any of the turtles used.

The results showed that the activity of serum ALT and AST significantly increased in turtles treated with 500 and 1000 mg kg⁻¹ concentrations, while no changes were found at 250 mg kg⁻¹ (Fig. 1). AST and ALT are cytoplasmic enzymes present mainly but not exclusively in the liver and which play an essential role in the synthesis of glutamate from α -ketoglutarate (Ellinger et al. 2011). The increased activity of AST and ALT in the serum of turtles may be indicative of tissue damage or inflammation affecting cell membranes and leading to the release of these enzymes to the bloodstream (Sinaei and Zare 2019). Since the cell membrane is permeable to α -ketoglutarate, the activity of the AST and ALT can exert a significant effect on intracellular glutamate levels (Banaee et al. 2020). As glutamate is one of the three primary amino acids in intracellular glutathione synthesis, its reduction can decrease glutathione levels and reduce the antioxidant capacity of cells. An increase in AST and ALT activities has also been reported in the serum of tilapia (*Oreochromis niloticus*) after exposure to 100 mg L⁻¹ of MPs for 15 days (Hamed et al. 2019). Also, Fazio et al. (2012) evidenced that increased or decreased activity of AST and ALT in the serum of turtles is indicative of disease and histopathological damage.

On the contrary, the GGT activity decreased significantly in the serum of turtles exposed to 500 and 1000 mg kg⁻¹ MPs compared to the control group (Fig. 1). GGT activity is essential to degradation of the glutathionylated xenobiotics in the hepatocytes (Koenig and Seneff 2015). After advancing the previously mentioned reaction, changes in the shape of the GGT active site may lead to its inactivation. The inactivated enzyme is then broken down, and its metabolites are eventually excreted in the bile (Fornaciari et al. 2014). Therefore, a decrease in the GGT activity in the turtles exposed to MPs may result in the collapse of the detoxification process. In fact, decreased GGT activity may be due to increased bile secretion (Li and Chiang 2014). GGT is a membrane enzyme that contributes to the degradation of extracellular glutathione to glutamate, cysteine, and glycine (Ellinger et al. 2011). In this sense, GGT plays a vital role in intracellular glutathione regeneration by providing essential amino acids for intracellular glutathione biosynthesis (Bradley 2016; Akaydin et al. 2020). Thus, a decrease in GGT activity may also contribute to a

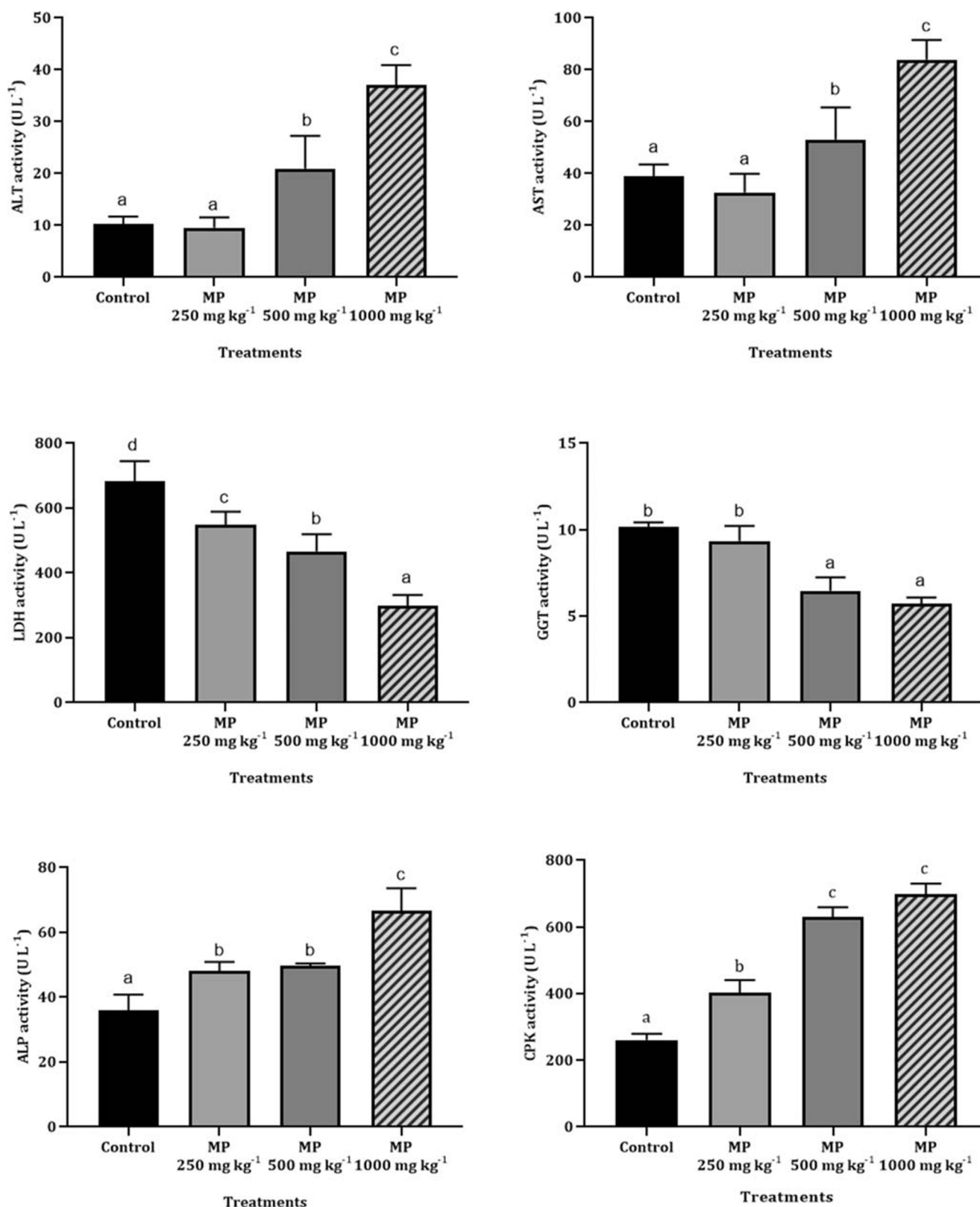


Fig. 1 The change of enzyme activities in the serum of *E. orbicularis* after oral exposure to different concentrations of MPs for 30 days. Data are displayed as mean ± SD. Significant changes between amounts as compared with the reference group are shown by different letters ($P < 0.05$)

reduction in the capacity of intracellular glutathione biosynthesis (Bradley 2016; Elnozahi et al. 2020). Furthermore,

GGT activity is essential for the maintenance of cysteine in the body. Cysteine is one of the three sulfur-containing amino

acids in the body of organisms and due to its ability to participate in oxidation/reduction reactions can act as a radical scavenger neutralizing O_2^- and H_2O_2 radicals (Kim et al. 2020). Consequently, a decrease in GGT activity can also affect cellular antioxidant capacity (Bradley 2016; Hatami et al. 2019).

LDH activity in the serum of turtles exposed to different MPs concentrations was significantly lower than the control group (Fig. 1). LDH catalyzes the reversible conversion of pyruvate to lactate using NADH and is involved in the anaerobic metabolism of glucose when oxygen supply is limited (Adeva-Anday et al. 2014). In this sense, LDH is an indicator of hypoxia and mitochondrial oxidative function. Moreover, increases in extracellular LDH activity may be indicative of cell death or necrosis (Maes et al. 2015). However, some chemical compounds such as plasticizers may alter the LDH activity since these compounds may inactivate LDH by interacting with its catalytic domain (Kendig and Tarloff 2007). Tangkosakul et al. (2009) also showed that oxidation of LDH by reactive oxygen species (ROS) could lead to LDH inactivation. Thus, the decrease in LDH activity in the blood of turtles may be due to LDH inactivation by plasticizers or free radical overproduction.

The ALP activity in turtles treated with different doses of MPs was considerably higher than the reference group (Fig. 1). Turtles exposed to 1000 mg kg⁻¹ MPs showed the highest activity for ALP in the serum. ALP is a glycoprotein enzyme that binds to the cell membrane (Banaee 2020). Therefore, cytoplasmic membrane damage can release ALP into the blood. Some of the underlying reasons leading to higher ALP activity are deficiency of calcium, zinc, vitamin D, and protein, as well as liver damage by xenobiotics (Sun et al. 2011; Banaee 2020). In accordance with the obtained results, it has been evidenced that MP exposure elevated blood levels of ALP in tilapia (*O. niloticus*) (Hamed et al. 2019).

CPK is a wide distributed enzyme found in significant amounts in muscles, heart tissue, lungs, kidney, and brain of turtles and is released to the bloodstream after damage to cells (Perrault et al. 2017; Pagano et al., 2019). The present results evidenced a significant increase in CPK activity level in all MP-treated turtles compared with the control group (Fig. 1). A high CPK level can also indicate oxidative stress or damage to muscles, kidney, and lungs (Mc Anulty et al., 2007; Luckoor, Salehi and Kunadu, 2017). Espinosa et al. (2017) also found that polyvinyl chloride (PVC-MPs) caused an increase in the activity of CPK in gilthead seabream (*Sparus aurata* L.). They showed that the increase in CPK in fish exposed to PVC-MPs could be due to damage to muscles and kidneys (Espinosa et al., 2017).

The intake of MPs in turtles through diet resulted in a significant decrease in total proteins and globulins (Fig. 2). Similarly, the albumin and total immunoglobulin levels in serum significantly decreased after exposure to 500 and

1000 mg kg⁻¹, but not in the 250 mg kg⁻¹ group (Fig. 2). Reduced levels of total protein, albumin, and globulin may be related to malnutrition, gastrointestinal dysfunction, and liver and kidney dysfunction (Novillo, Raga and Tomás, 2020) in the turtles exposed to MPs. Specifically, the increase in transaminase levels suggests the probability of liver damage that may alter its ability to synthesize proteins in a normal way. Zhao et al. (2020) showed that exposure zebrafish to MPs could lead to malnutrition. Moreover, decreased immunoglobulin levels may increase the risk of pathogen-induced infections in MP-exposed turtles. Ju et al. (2020) found that MPs may be transferred to vital organs by binding to albumin as a carrier biomolecule. Therefore, a decrease in albumin may have reduced the distribution rate and bioaccumulation of MPs in turtles. Furthermore, MPs could affect the physiological function of proteins by destroying their structure (Ju et al., 2020). Camacho et al. (2013) and Komoroske et al. (2011) reported that a decrease in the total protein, albumin, and globulin levels was attributed to liver damage in turtles exposed to xenobiotics.

Cholesterol levels in serum of turtles exposed to 500 and 1000 mg kg⁻¹ MPs were significantly higher, compared with the control group (Fig. 3). A significant portion of blood cholesterol is related to the lipid absorbed through food (Cohen, 2008). Therefore, impaired high-density lipoprotein (HDL) biosynthesis and decreased cholesterol excretion through bile secretions may lead to increased blood cholesterol. Similarly, a significant increase in total cholesterol was observed in the hemolymph of the red-claw crayfish (*Cherax quadricarinatus*) after 21 days of exposure to MPs (Chen et al., 2020). Previous studies by Hamed et al. (2019) and Banaee et al. (2019) also showed that MP intake caused an increase in cholesterol levels in the blood of *O. niloticus* and *C. carpio*, respectively. Serum triglyceride levels significantly increased in turtles treated with 1000 mg kg⁻¹ MPs (Fig. 3). Prolonged cellular hypoxia can lead to increased expression of genes involved in lipid metabolism and increase the lipolysis (Li et al., 2018). Therefore, an increase in triglycerides may indicate a disturbance in lipid metabolism in turtles exposed to MPs. Also, Sussarellu et al. (2016) found that exposure of aquatic organisms to MPs can decrease gene expression of enzymes involved in the oxidation of fatty acids and disrupt of fatty acid metabolism. Increased levels of triglycerides in the blood may be used to produce energy to reduce the stress after exposure to MPs. The same results were reported in the tropical freshwater crayfish, *Cherax quadricarinatus*, after 21 days of exposure to MPs (Chen et al., 2020). Exposure to 500 and 1000 mg MPs also resulted in increased glucose levels (Fig. 3). Elevated glucose levels are caused by the breakdown of hepatic glycogen to provide energy to counteract the toxic effect of plasticizers (Rui, 2014; Rezaei Shadegan and Banaee, 2018). Corticosteroids can control changes in glucose levels under different physiological conditions

(Jessop and Hamann, 2004). Therefore, the increase in blood glucose may have been due to elevated levels of corticosteroids in MP-exposed turtles. Also, raised blood glucose may also favor the conversion of excess glucose to triglycerides, helping to increase its circulating levels. Hamed et al. (2019) found that exposure to MPs caused hyperglycemia in the blood of tilapia (*O. niloticus*).

Feeding the turtles on a diet containing 500 and 1000 mg kg⁻¹ MPs resulted in a significant increase in serum creatinine and urea levels (Fig. 4). Urea is the nitrogen metabolite produced by the breakdown of proteins in the animal body as a mechanism to reduce nitrogen toxicity. As blood passes through Bowman's capsule, glomeruli filter out urea and excrete it in the urine (Korrapati and Mehendale, 2014). Therefore, the amount of urea in the proximal tubule of the nephrons determines the glomerular filtration rate. Changes in blood urea concentration may be a good indicator to assess nephron function (Perrault et al. 2017). The results of this

study indicated that exposure to MPs can lead to renal nephron damage and disrupt the glomerular filtration rate in turtles. Hamed et al. (2019) showed that the increase in the urea level in the blood of tilapia (*O. niloticus*) exposed to MPs might have been due to impaired renal function. Also, Tomizawa et al. (2015) found that histopathological damage accompanied with bleeding in the gastrointestinal tract can increase the urea levels (Tomizawa et al. 2015). Furthermore, increased urea in the blood may be due to the effect of MPs on the main metabolic pathways involved in the urea cycle (Jin et al. 2019). Creatinine is the most critical metabolic product produced by the muscles and excreted by the kidneys. If the kidneys fail to function correctly, the creatinine levels in the blood will be elevated. Equally, blood creatinine contents, like elevated urea levels, are a suitable diagnostic bio-indicator for assessing the glomerular filtration rate. Therefore, the increase in creatinine in the blood of MP-exposed turtles may be due to an alteration in the glomerular filtration rate. Hamed et al.

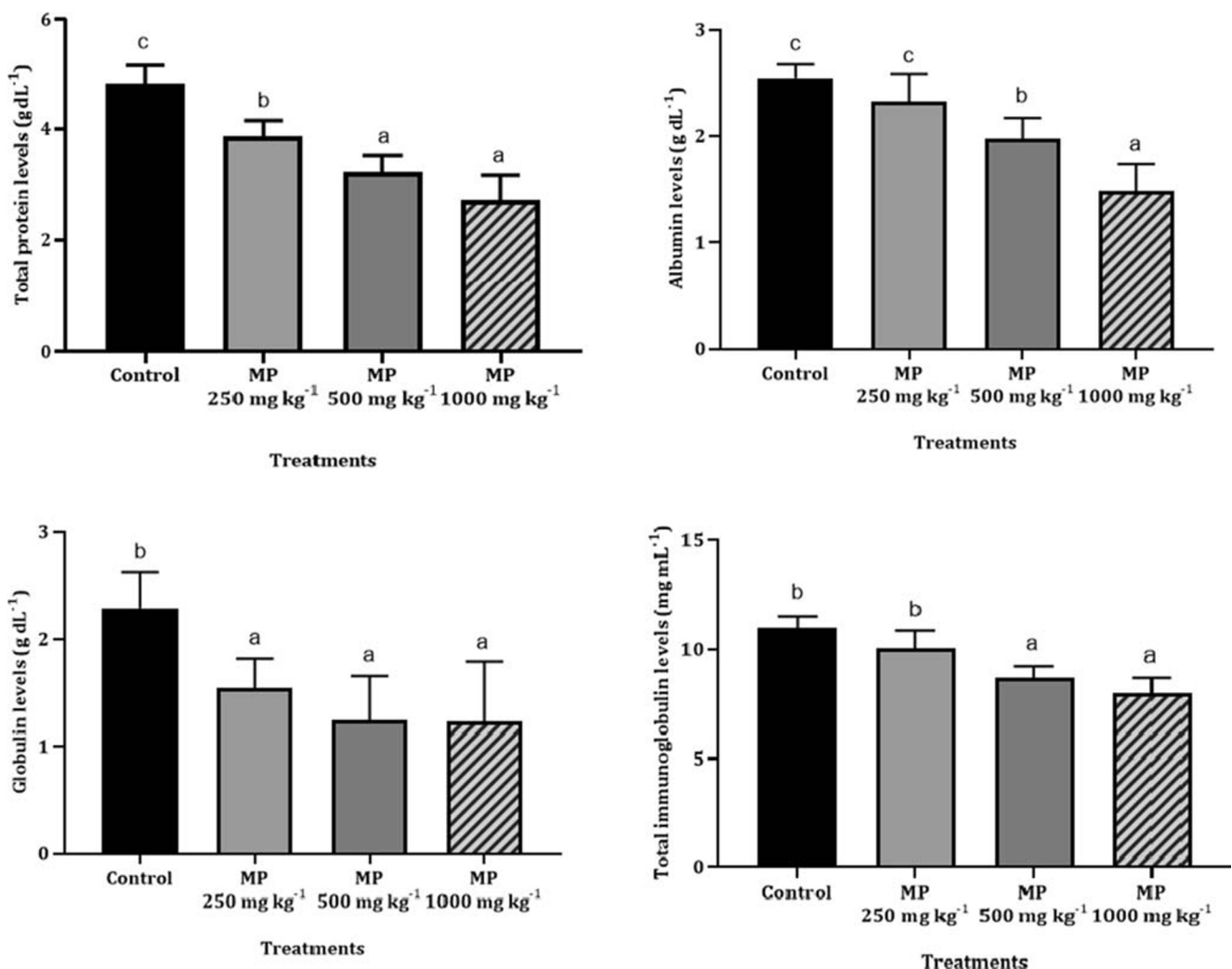


Fig. 2 The change of total protein, albumin, globulins, and total immunoglobulin levels in the serum of *E. orbicularis* after oral exposure to different concentrations of MPs for 30 days. Data are

displayed as mean $\pm$ SD. Significant changes between amounts as compared with the reference group are shown by different letters ($P < 0.05$)

(2019) found that exposure of tilapia (*O. niloticus*) to MPs resulted in enhanced creatinine levels.

The results indicated that blood calcium levels in turtles treated with 500 and 100 mg MP per kg of food were significantly higher than the control animals (Fig. 5). Furthermore, elevated blood calcium levels can increase the rate of calcium deposition in the kidneys, weaken bones stability, and impair the heart and brain functions of MP-exposed turtles. Elevated calcium levels could derive from the increased activity of the parathyroid glands and the appearance of hyperparathyroidism (Lunn et al. 2010). Following an increase in the level of parathyroid hormone in the blood, the calcium, phosphate, and vitamin D homeostasis are disrupted (Yuen et al. 2016). The relationship between calcium and phosphorus in the blood is reversed, meaning that as calcium levels increase, the concentration of phosphorus decreases (Moe, 2008). Feeding *E. orbicularis* with diets containing 500 and 1000 mg kg⁻¹ MPs significantly reduced phosphorus levels in the bloodstream (Fig. 5). This decrease could come from a state of malnutrition, an alteration in the absorption/excretion

of phosphorus in the intestines, and a higher concentration of calcium in the blood of the turtles. The presence of MPs could target kidneys increasing rate of phosphorus excretion and decreased the intestine capability to absorb this compound leading to reduced levels in the blood (Vervloet and Ballegooijen 2018).

Magnesium is one of the major blood electrolytes involved in the regulation of cell function, nerve conduction, and the activity of over 300 enzymes involved in processes such as glycolysis, Krebs cycle, beta oxidation, and ion transfer from the cell membrane (Uwitonze and Razzaque 2018; Pagano et al. 2019). Exposure to MPs through the diet significantly reduced magnesium in the blood of pond turtles probably associated with disturbances in the intestinal absorption (Fig. 5). A low magnesium level in blood could follow a disturbance in the digestive system, renal and hepatic failure (Ikee 2018; Liu et al. 2019). Furthermore, decreased magnesium in the blood can reduce the cardiovascular system activity (Ikee 2018).

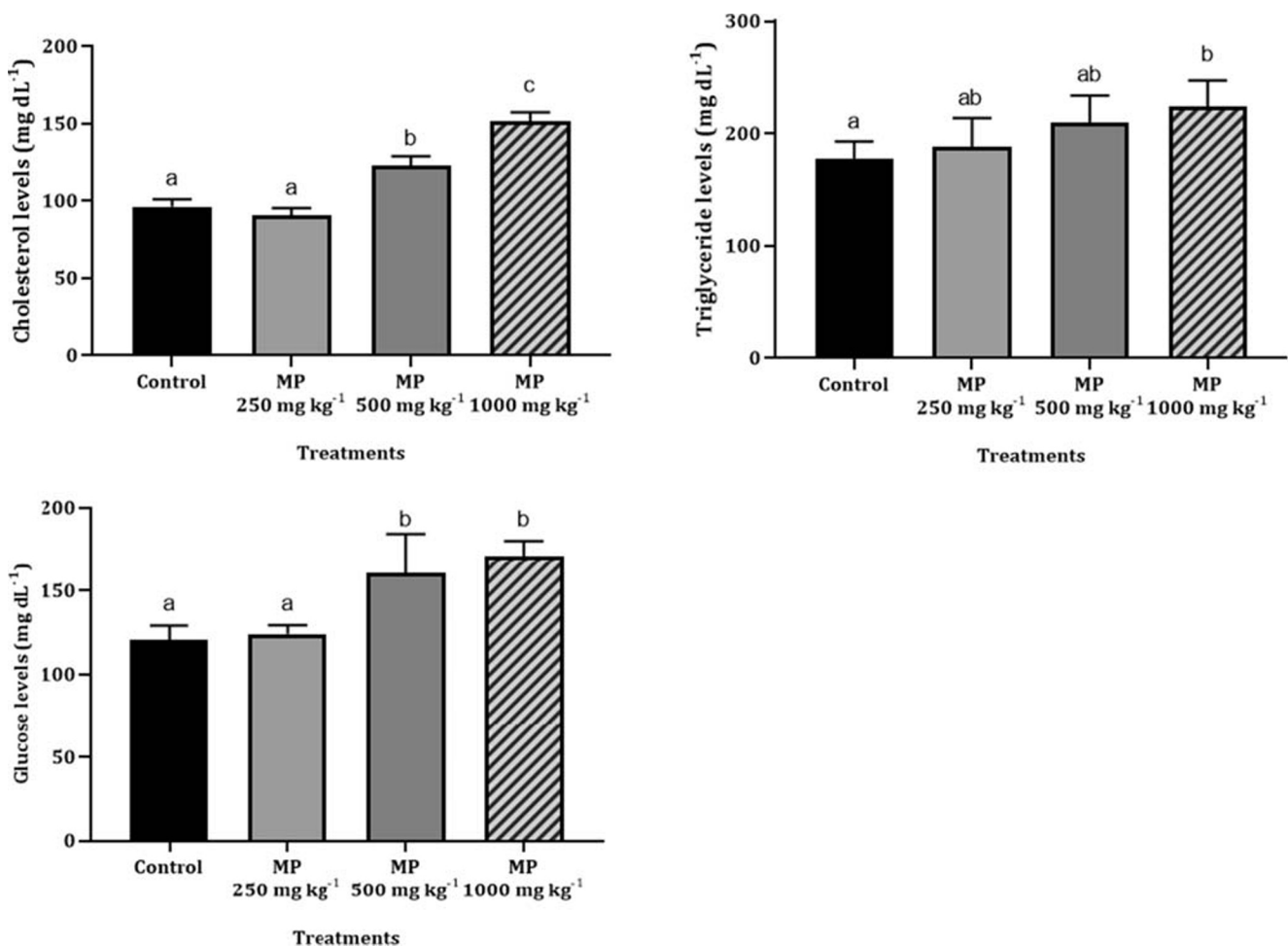


Fig. 3 The change of cholesterol, triglycerides, and glucose levels in the serum of *E. orbicularis* after oral exposure to different concentrations of MPs for 30 days. Data are displayed as mean $\pm$ SD. Significant changes

between amounts as compared with the reference group are shown by different letters ($P < 0.05$)

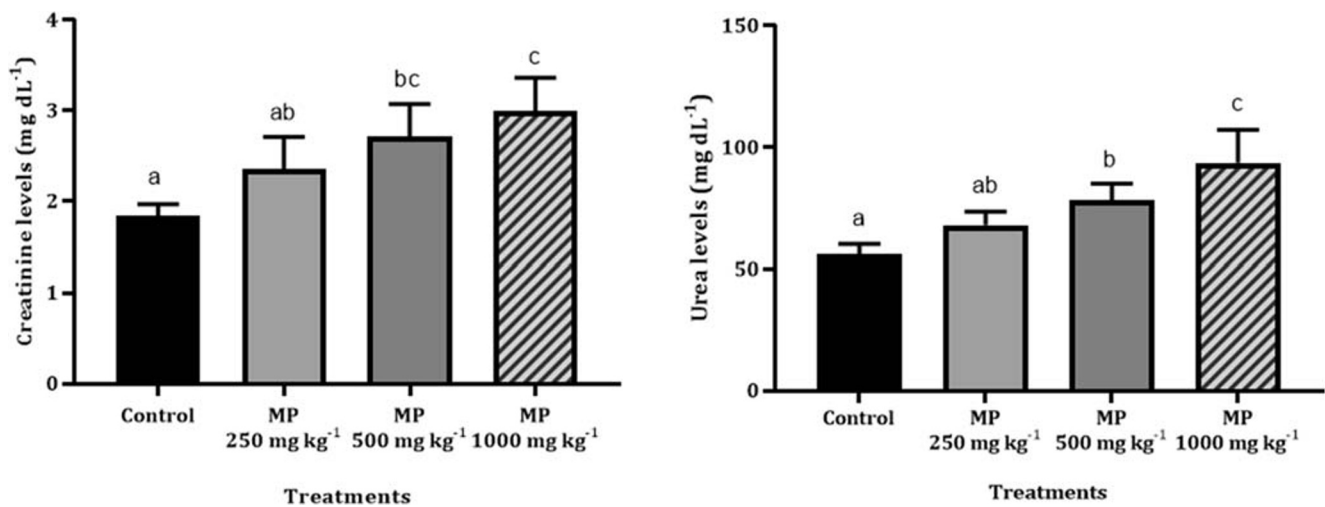


Fig. 4 The change of creatinine and urea levels in the serum of *E. orbicularis* after oral exposure to different concentrations of MPs for 30 days. Data are displayed as mean $\pm$ SD. Significant changes between amounts as compared with the reference group are shown by different letters ($P < 0.05$)

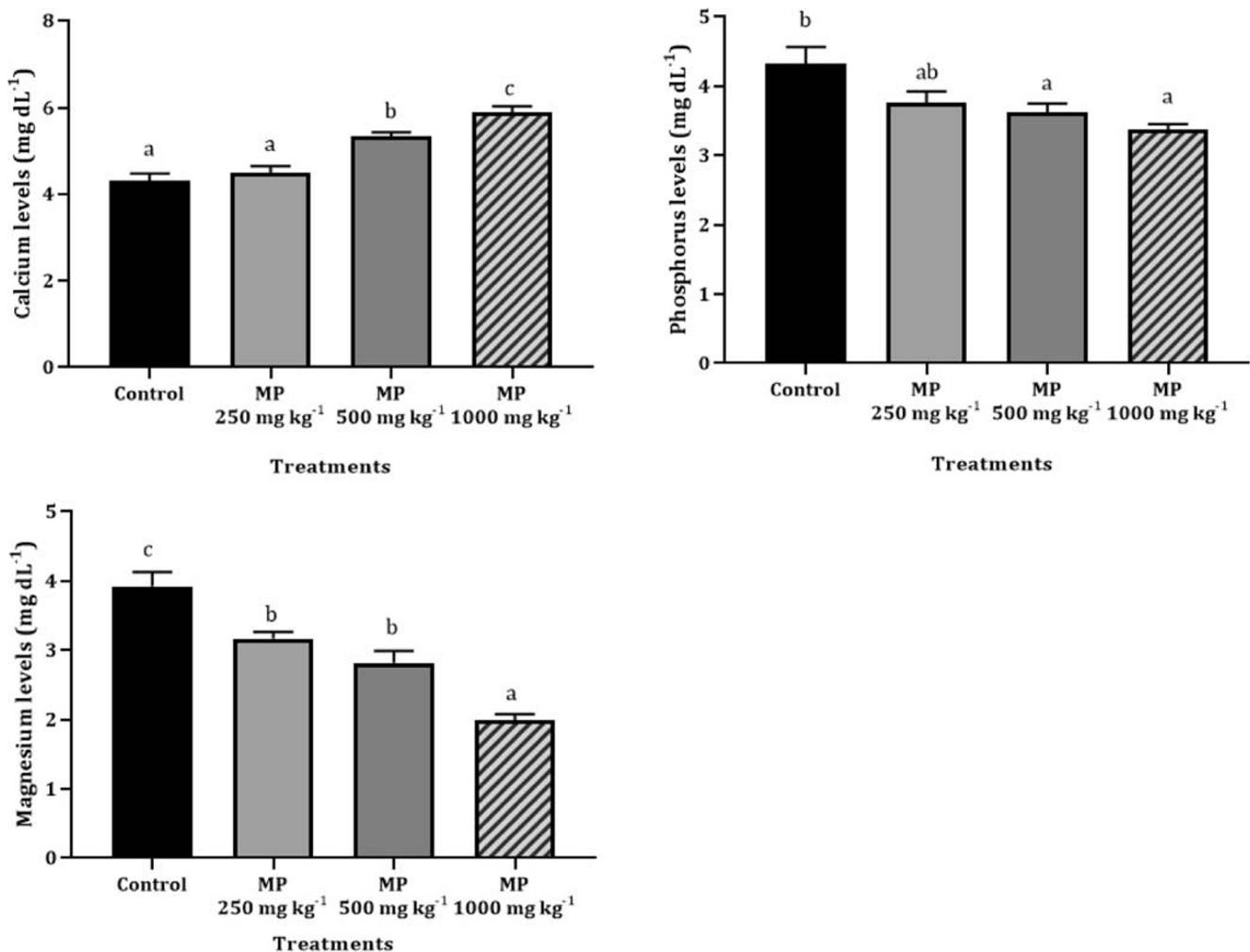


Fig. 5 The change of calcium, phosphorus, and magnesium levels in the serum of *E. orbicularis* after oral exposure to different concentrations of MPs for 30 days. Data are displayed as mean $\pm$ SD. Significant changes

between amounts as compared with the reference group are shown by different letters ($P < 0.05$)

Conclusion

The results of the present study indicated that exposure of *Emys orbicularis* to dietary MPs polyethylene PE100 altered all the blood biochemical parameters analyzed. The effects on the activity of blood enzymes, decreased protein synthesis, immunosuppression, and metabolic and electrolyte changes in the blood are indicative of changes in liver and kidney function. General homeostasis of the turtles may be related to ingestion of MPs. In conclusion, MPs are a significant threat to the health and survival of freshwater turtles; therefore, monitoring and reducing the entrance of these pollutants into aquatic ecosystems are essential.

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Authors' contributions Mahdi Banaee (Assistant Professor, Aquaculture and Ecotoxicology): Investigation, project administration, validation, formal analysis, and writing (original draft)

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Data availability The data that support the findings of this study are available from Behbahan Khatam Alanbia University of Technology, but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Behbahan Khatam Alanbia University of Technology.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval and consent to participate In this study, none of the authors used human beings as research subjects. International, national, and institutional guidelines for the care and use of animals were followed. Experimental protocols were done following the Iranian animal ethics framework under the supervision of the Iranian Society for the Prevention of Cruelty to Animals and Shiraz University Research Council (SURC973/46-2679-2).

Consent to publish In the present study, there was no individual person's data in any form (including any individual details, images, or videos).

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